

KUNAL PATIL

AI/ML Engineer

Mumbai, Maharashtra 400063 | kunalpatil970730@gmail.com | 8530397730

LinkedIn: linkedin.com/in/kp7730 | GitHub: github.com/kunalp02

PROFILE

AI/ML Engineer with 1+ years of experience building production AI systems, LLM agents, RAG pipelines, and MCP integrations using Python, FastAPI, LangChain, and Ollama. Delivered enterprise solutions at TCS including natural-language SQL tools, observability MCP servers, and AI chat platforms. Strong in system integration, RBAC/security, Docker, monitoring, production troubleshooting, and cross-team collaboration. Seeking to deliver customer-facing AI and hybrid IT solutions in onsite consulting environments.

PROFESSIONAL EXPERIENCE

Software Engineer | Tata Consultancy Services - Mumbai, India | Jan 2025 - Present

- Built an NL-to-SQL assistant using Python, LangChain, Ollama, and ChromaDB, enabling business and ops teams to generate reports via conversational queries and reducing manual SQL effort.
- Developed a Grafana MCP Server exposing metrics and logs as MCP tools, enabling AI assistants to query observability data through standardized tool calls across applications.
- Customized OpenWebUI with locally hosted Ollama models, making AI accessible to developers and business users through a production-ready chat interface.
- Set up and maintained monitoring dashboards (Grafana Alloy, Prometheus, Loki) for 5 production trading applications; partnered with dev and support teams on incident troubleshooting and system stability.
- Automated daily operational workflows (PRE, POST, EOD, INTRADAY) using Jenkins Groovy pipelines, saving several hours of manual effort per day for the support team.
- Implemented Keycloak-based Active Directory authentication and mTLS via NGINX for secure service-to-service communication in internal platforms.

TECHNICAL SKILLS

AI/ML: LangChain, LangGraph, RAG, MCP (Model Context Protocol), Ollama, Qdrant, ChromaDB, PGVector, AWS Bedrock, multi-agent workflows, Text-to-SQL

Backend: Python, FastAPI, Django, REST APIs, SQL, JWT, OAuth 2.1, RBAC, async job processing

DevOps & Cloud: Docker, Jenkins, GitHub Actions, AWS (EC2, S3), Linux, Ansible, Terraform, NGINX, mTLS

Data & Observability: PostgreSQL, MySQL, MongoDB, Redis, Prometheus, Grafana Alloy, Loki, LangSmith

Frontend & Tools: React.js, Git, Postman

PROJECTS

Orchestrator - Multi-Agent AI Platform | Python, FastAPI, LangChain, Qdrant, Redis, PostgreSQL, React, Docker, MCP | github.com/kunalp02/Accuknox

- Built a multi-tenant SaaS orchestrator for agent workflows, async execution, and knowledge-base RAG using FastAPI, ARQ workers, Qdrant, and an OpenAI-compatible LLM gateway (Ollama/Bifrost).
- Implemented multi-agent workflows (agent, supervisor, branch, parallel, human-in-the-loop) with async invoke, run tracking, and SSE event streaming.
- Integrated MCP HTTP connections for external tool discovery and agent tool binding in production-style agent execution loops.
- Designed RBAC, scoped API keys, org isolation, and encrypted secret storage; containerized stack with Docker Compose and CI via GitHub Actions.

MCP AuthN/AuthZ - Secure MCP Client & Server | Python, FastAPI, OAuth 2.1, RFC 9728/8693, JWT, RBAC | github.com/kunalp02/Accuknox

- Built standalone MCP client and server with inbound JSON-RPC (tools/list, tools/call) and outbound HTTP MCP client for secure tool invocation.
- Implemented authentication and authorization: JWT, scoped API keys, RBAC (owner/admin/builder/viewer), tool allowlists, and audit logging at invoke time.
- Added industry-standard OAuth flows: RFC 9728 protected-resource metadata, PKCE, token refresh, and OBO token exchange (RFC 8693 + Microsoft Entra format).
- Wrote automated tests (pytest) for RBAC, policy enforcement, and OBO exchange; extends production MCP and Keycloak auth work from TCS.

EDUCATION

B.E. Computer Engineering | SSBT COET, Jalgaon | 2020 - 2024 | CGPA: 8.84/10

CERTIFICATIONS

Docker Foundations Professional Certificate (Docker) | Technical Content Engineer Certificate (GeeksForGeeks)